Measure, Gate, Train: Budget-Aware Self-Configuring Neural Architecture Search for Unseen Data

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1 Motivation

The NAS Unseen-Data Challenge (AutoML 2026) evaluates a fully automated pipeline on *hidden* classification datasets under a strict wall-clock budget per dataset. Inputs arrive as image-shaped tensors but range from natural and satellite images to *encoded* data (rendered text, Sudoku boards, chess positions) where the spatial prior of a CNN misleads. All decisions must be made at runtime, blind.

Our starting point is the winning method of the 2025 edition, EGNAS [1], a macro–micro search (depth, width, per-layer operation) over convolutional blocks. Reproducing it surfaced two systematic weaknesses. First, rankings from short training runs are unreliable: operations that lead after a few epochs often lose after full training, so fine-grained (micro) operation search tends to *harm* final accuracy. Second, the space is confined to one architecture family: in our preliminary experiments a *fixed* 2-D dilated-convolution backbone in the style of temporal convolutional networks [3] clearly outperformed the fully searched CNN baseline on most datasets, while plain CNNs and randomly sampled architectures won precisely on the encoded datasets, in line with random search being a strong NAS baseline [2, 4]. The leverage lies in choosing the right architecture *family* per dataset, not in fine-grained operation search.

2 Research Questions

RQ1. Can macro–micro NAS [1] be generalised across heterogeneous architecture families (plain CNNs, 2-D dilated and 1-D temporal convolution networks), making families *comparable* within a single time-budgeted search stage despite the known unreliability of low-fidelity rankings [5]? **RQ2.** Can cheap, training-free measurements of the raw data decide at runtime which architectural prior fits an unseen dataset? **RQ3.** How can harmful data augmentation be detected at runtime, given that augmentation acts as regularisation (it costs accuracy early and pays late) and that, as we observe, augmentation decisions do not transfer between architecture families?

3 Approach

We propose a *measure–gate–train* pipeline (figure on page 2). (1) *Measure*: a fingerprint computed directly from the raw input tensors, with no training involved. The mean correlation of neighbouring pixels separates the two data families with a wide margin: between -0.05 and 0.07 on the four encoded public datasets versus 0.38 to 1.00 on all others. (2) *Macro stage*: for image-like data the dilated backbone is selected directly. Only encoded data pays for a brief bake-off of all families at matched wall-clock. The leader is adopted only if it beats the robust default by more than the measured ranking noise, otherwise the pipeline *abstains* to the default, which makes every decision bounded-downside. (3) *Augmentation probe*: a two-arm A/B test run *after* the family choice with arms of the chosen family. It deviates from the default only on a clear margin (RQ3). (4) *Micro stage*: depth search anchored on an incumbent default. A deviation must win significantly *and* survive verification by re-training at a longer horizon. (5) A *random-search arm* over a wide operation vocabulary runs only on encoded data and only if budget remains [2]. (6) *Train to budget*: the winner trains on the full remaining time with a cosine learning-rate schedule [6] sized to reach zero exactly when the budget ends. All stage budgets are fractions of the clock, so the pipeline degrades gracefully from many hours down to minutes.

4 Required Steps

(1) Reproduce the EGNAS baseline and probe candidate architecture families at full budget (done). (2) Validate each decision mechanism in isolation by replaying it against known full-budget outcomes (gate reliability, fingerprint separation, augmentation transfer). (3) Integrate the pipeline and verify all execution paths, including the low-budget collapse behaviour. (4) Full multi-seed evaluation. (5) A budget-ladder study (0.5h to 8h) measuring which decision stages pay at which budget. (6) A generalisation test on untouched external datasets (CIFAR-10/100, EMNIST, MedMNIST, spectrograms). (7) Packaging and stress tests.

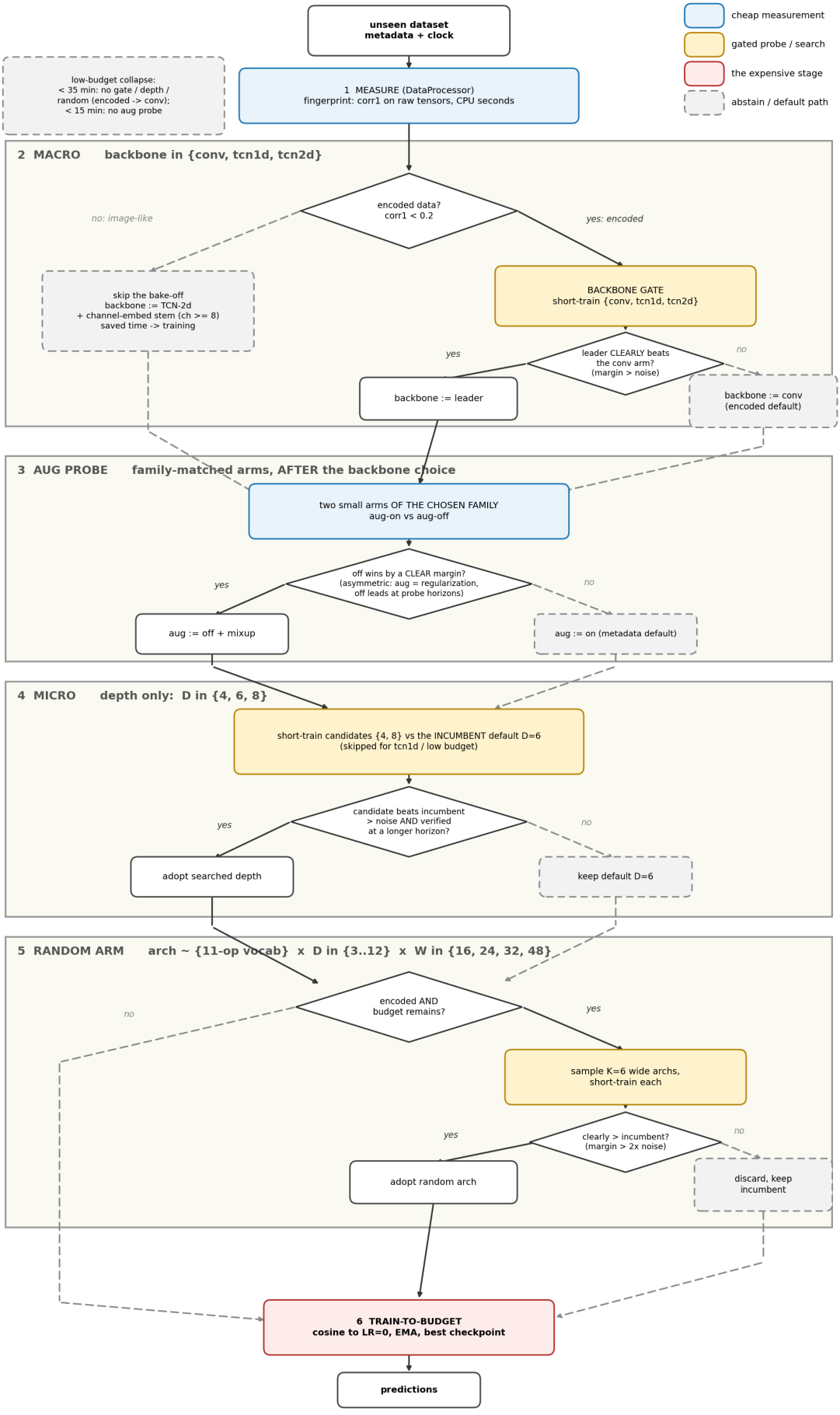
5 Evaluation Strategy

Primary evaluation uses the 13 public competition datasets with the official adjusted score (accuracy versus a per-dataset benchmark, capped), at least three seeds for final configurations. Baselines: the reproduced EGNAS search, each fixed family, and a best-family oracle bounding the achievable routing gain. Every stage writes its measurements, margins, and choices to a per-dataset decision record, so gains and losses are attributable to individual decisions. Robustness is evaluated across budgets (30 minutes to 12 hours), on the external suite, and against published per-dataset competition bests [2]. A first complete run of the pipeline (one seed, 2 h per dataset) totals an adjusted score of **+62.5** on the 13 public datasets, versus +48.2 for the searched-CNN baseline and +63.2 for the family oracle (figure on p. 3).

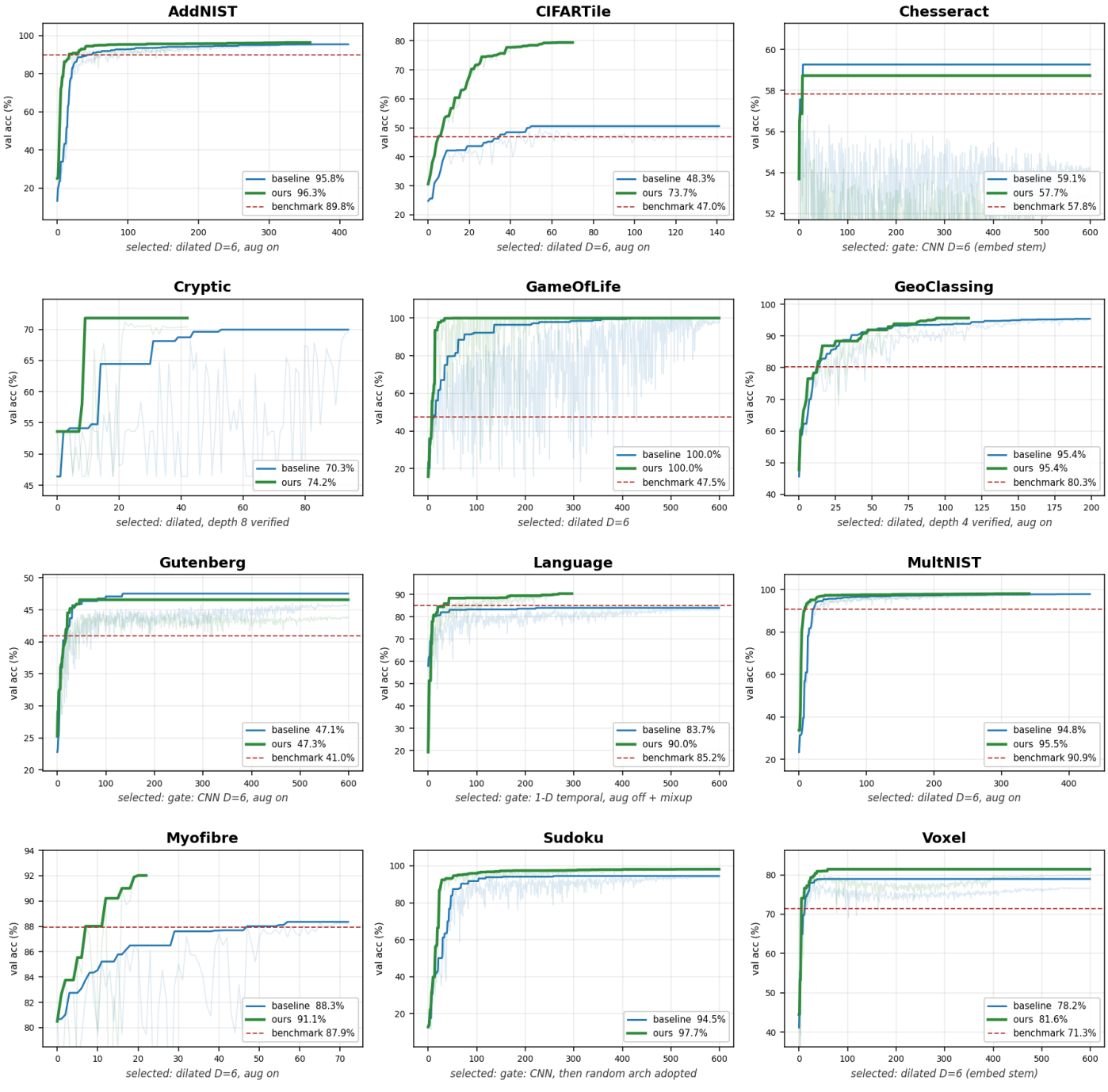
6 Potential Extensions

If time allows, the pipeline's ambition will scale with the budget: additional families (e.g. recurrent networks, which need longer probes to rank fairly) join the bake-off only at large budgets, and when families are statistically tied, the surplus is spent on a heterogeneous ensemble, kept only if it beats the best single model on validation. In parallel we are creating competition-style datasets, designed blind to the pipeline (e.g. QR codes with partially covered regions, encoded board games), as a further generalisation test.

The measure-gate-train decision pipeline (run once per unseen dataset)



First results: our pipeline vs the searched-CNN baseline (13 public datasets)



Validation curves over training epochs (2 h per dataset, one seed).

ours: the measure-gate-train pipeline; final test accuracy in the legend, the configuration it selected below each panel.

baseline: the reproduced searched-CNN approach.

Totals (adjusted score, 13 sets):
  baseline +48.2
  ours +62.5
  family oracle +63.2

Sudoku and Cryptic have no published benchmark (0.0).

References

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[3] S. Bai, J. Z. Kolter, V. Koltun. *An Empirical Evaluation of Generic Convolutional and Recurrent Networks for Sequence Modeling*. arXiv:1803.01271, 2018.

[4] L. Li, A. Talwalkar. *Random Search and Reproducibility for Neural Architecture Search*. UAI 2019.

[5] C. White, A. Zela, B. Ru, Y. Liu, F. Hutter. *How Powerful are Performance Predictors in Neural Architecture Search?* NeurIPS 2021.

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